

# NE 5350 – Nuclear Criticality Safety Homework 1

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# 1 Mayak 1958 Criticality Accident

Using the concepts of exponential power growth and reactivity feedback discussed in class, describe the Mayak 1958 criticality accident.

In a steady-state condition, in which a system remains in 'static reactivity', describes the state in which the 90% enriched uranyl nitrate solution remained in the cylinder, sub-critical as the reactivity was less than zero.

$$\rho = \frac{k_{eff} - 1}{k_{eff}} < 0 \quad (1)$$

Reactivity ( $\rho$ ) is dependent on the neutron population; as the neutron population increases within a system,  $\rho$  increases, and the opposite is also true when  $\rho$  decreases, the neutron population decreases. Within critical and sub-critical systems, the neutron population is proportional to power.

During operations after an experiment, the experimentalists were emptying the tank with six-liter bottles, being of favorable geometry. They began to deviate from procedure, as this work was tedious. After assuming the remaining quantity of uranyl nitrate were presumably sub-critical, the experimentalists decided to unbolt and tilt the tank over to fill the bottles faster [1]. As they tilt the tank over, as depicted in Figure 1, the solution inside favored one side of the tank and resembled the geometry of a sphere -reducing neutron leakage- and causing a prompt criticality event due to the exponential power growth of the surging sequential nuclear generations and reactivity feedback.



Figure 1: Solution Sitting Flat in Cylinder to Becoming Sphere-like

## 1.1 Feedback Effects

Once the favorable geometry (of the partially filled cylindrical tank) went out the window, which kept the solution distributed across a wide surface area, the uranyl nitrate solution went prompt super-critical. The reactivity of the system drastically increased due to the sudden change in geometry, exponentially increasing the neutron population in less than a second, and producing a bright flash of Cherenkov light. In this process, the solution explosively shot toward the ceiling, as much of the material changed phase to a gaseous form due to the sudden increase in power born from the criticality incident. This material change decreased reactivity, as the uranyl nitrate was no longer compactly contained in the tilted cylinder. The ejection of material fundamentally changed the system in a way that a re-criticality event was no longer possible. In the process of thermal expansion, prior to the ejection of fissile material, neutron leakage

increased as reactivity decreased.

## 1.2 Exponential Growth

Consider the production and "destruction" of neutrons and delayed neutron precursors (DNPs), which have been provided during the lecture, as an approximation of a zero-dimensional system:

$$\frac{d}{dt}n(t) = \frac{\rho(t) - \beta}{\Lambda}n(t) + \sum_{d=1}^{N_D} \lambda_d C_d(t) \quad (2)$$

$$\frac{d}{dt}C_d(t) = \frac{\beta_d}{\Lambda}n(t) - \lambda_d C_d(t) \quad (3)$$

where  $n(t)$  is the neutron population,  $\rho$  is reactivity,  $\beta_d$  is the effective delayed neutron fraction,  $\lambda_d$  is the decay constant [ $s^{-1}$ ],  $\Lambda$  is the generation time, and  $C_d$  is the DNP concentration. As reactivity ( $\rho$ ) increases, due to a change of the geometry, so does the neutron population, that is proportional to power. In an analysis using OpenFOAM and Serpent2, Vega et al. determined that tilting the cylinder resulted in a reactivity change of  $\Delta\rho_{geometry} = \$27.47$ , with  $k_{eff}$  ranging from 0.9142 to more than 1.17 prior to becoming subcritical after the material ejection [2]. As discussed during Lecture 2, an insertion of \$0.50 produces exponential growth in the neutron population, let alone the insertion of \$27.47, and will continue to grow exponentially until the system fundamentally changes.

## 2 Demon Core Criticality Accidents

In the first “Demon Core” criticality accident, the operator remained in the room after the accident thereby significantly increasing the dose that he received. In the second “Demon Core” criticality accident, the operator described physically separating the two hemi-spheres though the power excursion had already been terminated by reactivity feedback effects. For both of these criticality safety accidents, describe how the physics of reactivity feedback could have been used to modify the experiments to make them safer.

With both criticality accidents, worker dose could have been prevented if the experiments were conducted remotely, like within a hot cell. However, these two accidents occurred in the 1940s, so experimentalists were constrained by limitations in procedures to carry out criticality experiments or the necessary foresight to do so. Each accident occurred after a slip in the fine motor skills of the worker. The following subsections will discuss how the experiments could have been made safer using reactivity feedback, despite the experimentalists being in the same room.

### 2.1 Demon Core I

The “demon core” was a 6.2 kg sphere of plutonium, composed of two hemispheres [1]. The critical assembly was being created – by hand – by stacking 4.4 kg tungsten carbide bricks compactly around the sphere. The purpose of the tungsten carbide bricks was to reflect escaping neutrons back into the plutonium core, increasing reactivity. With the experimentalist, slipping his hand and dropping the final reflecting tungsten carbide brick on the assembly, he stayed in the room after the sphere went prompt supercritical. Before modifications to the experiment is explored, it should be stated that the experimentalist should have fled the scene to reduce the radiation dose exposure.

Tungsten carbide is dense, appropriate for a reflector material. As the bricks were arranged compactly around the core, it may have been safer for the distance to increase between the plutonium core and the surrounding tungsten carbide bricks. Not only would this give the experimentalist more time to flee in the case of an accident, but would increase the distance for the gamma and neutron radiation to travel, reducing the effective dose on the worker.

As the experiment composed of simple components, the sphere and the surrounding bricks, this setup could have been assembled in a warehouse with cranes to lift and lower the bricks and sphere into place. Once the system became critical, in this hypothetical exercise, the worker in this scenario would still receive a dose, but to a lesser extent.

### 2.2 Demon Core II

In the second demon core criticality accident, the core hemispheres were encased in beryllium as a reflector material. After the first man died the previous year, first and foremost, I would not be in the same room as the 6.2 kg demon core of plutonium. To modify this demonstration, the shims should have been relied on to perform the criticality experiment, as opposed to the screwdriver separating the two hemispheres. The experimentalist did this to save on time, as he wanted to wrap things up before lunch. Furthermore, it should be stated that the demonstrator, after causing the prompt supercriticality, should have fled the room along with the others, rather than separating the hemispheres and collecting a substantial, lethal dose. The hemispheres being left together may have brought upon re-criticalities. Recall that a decrease in temperature and thermal expansion increases reactivity, and vice-versa.

Assume that there is additional material surrounding the beryllium shell of the demon core.

Within the hemispheres, suppose that there could be electric heaters in the material. Upon any transient conditions, the heaters would activate to increase the temperature within the hemispheres. Through the effect of Doppler Broadening, the resonance peaks would increase in the non-fissile nuclei of the shell, in which neutron absorption would be more likely to occur. This would fundamentally change the system, and drive down reactivity – provided that the heaters would stay on. However, this may prove to be ineffective if the heater coils fail to raise temperature enough to induce this effect in the material.

## References

- [1] McLaughlin, Thomas P., Monahan, Shean P., Pruvost, Norman L., Frolov, Vladimir V., Ryazanov, Boris G., Sviridov, Victor I., 2000 May. A Review of Criticality Accidents, 2000 Revision. Technical Report LA-13638, Los Alamos National Laboratory, Los Alamos, NM. URL <https://www.nrc.gov/docs/ML0037/ML003731912.pdf>
- [2] Vega, Richard M., Lane, Taylor K., Miller, John A., Schwes, Norman F., 2015 September. Reactivity Effects at the Mayak Production Association, January 2, 1958 Criticality Accident Using Serpent 2 and OpenFOAM. In: Proceedings of the International Conference on Nuclear Criticality Safety (ICNC 2015). American Nuclear Society, Charlotte, NC. URL <https://www.osti.gov/biblio/1255788>